Assignment 4

Q1. The number of continuous functions $f : [0,1] \to R$ satisfying

$$\int_0^1 f(x)\,dx \geq \frac{1}{3} + \int_0^1 f^2(x^2)\,dx \quad \text{--- i}$$

1) 0 2) 2 3) 1 4) ∞

Soln:

$$\int_0^1 f(x)\,dx \geq \frac{1}{3} + \int_0^1 f^2(x^2)\,dx$$

$$\int_0^1 f(x)\,dx \geq \int_0^1 x^2\,dx + \int_0^1 f^2(x^2)\,dx$$

Let $x = t^2$

$$\int_0^1 f(t^2)\,2t\,dt \geq \int_0^1 x^2\,dx + \int_0^1 f^2(x^2)\,dx$$

$$0 \geq \int_0^1 [f(x^2) - x]^2\,dx$$

$$f(x^2) = x \Rightarrow f(x) = \sqrt{x}.$$

Q2. If $\lim_{n\to\infty} \left((n+1)!^{\frac{1}{n+1}} - (n!)^{\frac{1}{n}} \right) = \frac{a}{b}$ then $\left[\frac{b}{a}\right]$ = ?

Soln: Take $y = (n+1)!^{\frac{1}{n+1}}$

$$\log y = \log (n+1)!^{\frac{1}{n+1}} = \frac{1}{n+1} (\log (n+1)!)$$

$$= \frac{\log(n+1) + \log n + \log(n-1) + \cdots + \log 1}{n+1}$$

Adding and subtracting $\log(n+1)$

$$= \frac{\log(n+1) + \log n + \cdots + \log 1}{n+1} - \log(n+1)$$

$$= \frac{\log(n+1) + \log n + \cdots + \log 1 - (n+1)\log(n+1)}{n+1} + \log(n+1)$$

$$= \frac{\log\left(\frac{n+1}{n+1}\right) + \log\left(\frac{n}{n+1}\right) + \cdots + \log\left(\frac{1}{n+1}\right)}{n+1}$$

$$= \frac{\left[\log 1 + \log\left(1 - \frac{1}{n+1}\right) + \log\left(1 - \frac{2}{n+1}\right) + \cdots + \log\left(1 - \frac{n}{n+1}\right)\right]}{n+1} + \log(n+1)$$

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using integration as a limit of a sum.

Taking $\frac{r}{n+1} = x$ where r values from 0 to n.

$= \int_0^1 \log(1-x)\,dx + \log(n+1)$

$= x\log(1-x) - \int x\left(-\frac{1}{1-x}\right)dx + \log(n+1)$

$= x\log(1-x) - \int dx + \int \frac{dx}{1-x} + \log(n+1)$

$= x\log(1-x) - x - \log(1-x) + \log(n+1)$

$= [(x-1)\log(1-x)]_0^1 - [x]_0^1 + \log(n+1)$

$= 0 - 1 + \log(n+1)$

$\log y = \log(n+1) - 1, \quad y = e^{\log(n+1)-1} = \frac{e^{\log(n+1)}}{e}$

$= \frac{n+1}{e}$

Similarly $(n!)^{\frac{1}{n!}} = \frac{n}{e}$

$(n+1)!^{\frac{1}{n+1}} - (n!)^{\frac{1}{n!}} = \frac{n+1}{e} - \frac{n}{e} = \frac{1}{e}$

Q3. If $f: (0, \frac{\pi}{2}) \to R^+$ be a real valued function satisfies the differential equation $x f'(x) = 1 + f(x)(x^2 f(x) - 1)$ and $f(\frac{\pi}{4}) = \frac{4}{\pi}$ and also $A = \int_0^{\pi/2} (\cos x) f(x)\,dx$ then which of the following is/are true?

(A) $\frac{2}{\pi} < A < 1$ (B) $1 < A < \frac{\pi}{2}$

(C) $\frac{7}{6} < A < \frac{\pi+6}{6}$ (D) $\frac{1+\sqrt{2}}{2} < A < \frac{\pi+2\sqrt{2}}{4}$

Soln:- $x f'(x) = 1 + f(x)(x^2 f(x) - 1)$

$x\frac{dy}{dx} = 1 + 1 + x^2y^2 - y$

$\frac{x\,dy + y\,dx}{dx} = 1 + x^2y^2, \quad \frac{d(xy)}{1+x^2y^2} = dx$

$\tan^{-1}(xy) = x + c, \quad \tan(x+c) = xy$

Since $f(\frac{\pi}{4}) = \frac{4}{\pi}$, $\tan x = xy$

$A = \int_0^{\pi/2} \cos x\, f(x)\,dx = \int_0^{\pi/2} \cos x \times \frac{\tan x}{x}\,dx$

$= \int_0^{\pi/2} \frac{\sin x}{x}\,dx,$

$\frac{2}{\pi} \le \frac{\sin x}{x} \le 1$, $\int_0^{\pi/2} \frac{2}{\pi}\,dx \le \int_0^{\pi/2} \frac{\sin x}{x}\,dx$

$1 \le A \le \frac{\pi}{2}$, $A = \int_0^{\pi/6} \frac{\sin x}{x}\,dx + \int_{\pi/6}^{\pi/2} \frac{\sin x}{x}\,dx$

$\int_0^{\pi/6} \frac{6}{\pi \times 2}\,dx + \int_{\pi/6}^{\pi/2} \frac{2\,dx}{\pi} \le A \le$

$\int_0^{\pi/6} 1\,dx + \int_{\pi/6}^{\pi/2} \frac{2\,dx}{\pi} \le A \le \int_0^{\pi/6} 1\,dx +$

$\Rightarrow \frac{1}{2} + \frac{2}{\pi}\left(\frac{2\pi}{6}\right) \le A \le \frac{\pi}{6} + \frac{3}{\pi}\left(\frac{2\pi}{6}\right)$ $\int_{\pi/6}^{\pi/2} \frac{6}{2\pi}\,dx$

$\frac{\pi}{6} \le A \le \frac{\pi+6}{6}$.

IIly $A = \int_0^{\pi/4} \frac{\sin x}{x}\,dx + \int_{\pi/4}^{\pi/2} \frac{\sin x}{x}\,dx$

$\int_0^{\pi/4} \frac{4}{\sqrt{2}\times\pi}\,dx + \int_{\pi/4}^{\pi/2} \frac{2}{\pi}\,dx \le A \le$

$\frac{1}{\sqrt{2}} + \frac{1}{2} \le A \le \frac{\pi}{4} + \frac{1}{\sqrt{2}}$, $\int_0^{\pi/4} 1\,dx + \int_{\pi/4}^{\pi/2} \frac{4}{\sqrt{2}\,\pi}\,dx$

$\frac{1+\sqrt{2}}{2} \le A \le \frac{1+2\sqrt{2}}{4}$.

Q4. $\lim_{n\to\infty} \left(\binom{n}{0}\binom{n}{1}\binom{n}{2}\cdots\binom{n}{n} \right)^{1/n(n+1)}$

(A) $\frac{1}{\sqrt{2}}$ (B) $\sqrt{2}$ (C) $\frac{1}{\sqrt{e}}$ (D) $\sqrt{e}$.

Q5. If $P = \lim_{n\to\infty} \prod_{r=1}^{n} \left(1 + \frac{1}{4r^2-1}\right)$ then $[37P]$ is (where $[.]$ is GIF)

(A) 55 (B) 57 (C) 58 (D) 59.

Q6. Let $g: N \to R^+$ is given by, $g(x) = \frac{1}{\sqrt{x}}$ and the value of $\left[\sum_{x=1}^{100} g(x)\right]$ is k, then the value of $\frac{k}{2}$ —— (where $[.]$ is GIF)

Q7. Let $S = \int_1^{e^2} \frac{\ln x}{\sqrt{x}}\,dx$ then which of the following is/are correct?

(A) $S < 2\left(e - \frac{1}{e}\right)$ (B) $S < (e-1)\left(2 + \frac{1}{\sqrt{e}}\right)$

(C) $S = \int_0^2 t\sqrt{e^t}\,dt$ (D) $3 < S < 5$.

Q8. If $T_n = \sum_{r=n}^{3n-1} \frac{r}{r^2+n^2}$, $S_n = \sum_{r=2n+1}^{3n} \frac{r}{r^2+n^2}$ (4)

then $\forall n \in \{1, 2, 3, \ldots\}$, which of the following is/are true?

(A) $T_n > \frac{1}{2}\log 2$ (B) $T_n < \frac{1}{2}\log 2$

(C) $S_n > \frac{1}{2}\log 2$ (D) $S_n < \frac{1}{2}\log 2$.

Q9. which is/are correct?

(A) $\int_0^\infty \frac{\sin t \cos t}{t}\,dt = \frac{\pi}{4}$ if $\int_0^\infty \frac{\sin t}{t}\,dt = \frac{\pi}{2}$

(B) $\int_1^k \ln x\,dx > \frac{1}{2}\ln k + \ln((k-1)!)$, $k \in N$

(C) $k! < e\sqrt{k}\left(\frac{k}{e}\right)^k$, $k \in N$

(D) If $a_n = \sum_{k=1}^{n} \frac{2^{k/n}}{n+\frac{1}{k}}$ then $\lim_{n\to\infty} a_n = \frac{1}{\log 2}$.

Q10. Let $I_1 = \int_0^{\pi/4} e^{x^2}\,dx$, $I_2 = \int_0^{\pi/4} e^{x^2}\sin x\,dx$,

$I_3 = \int_0^{\pi/4} e^{x^2}\cos x\,dx$, $I_4 = \int_0^{\pi/4} e^{x^2}\sin x\,dx$

then which of the following is/are true?

(A) $I_2 > I_4 > I_3$ (B) $I_2 \geq I_1 \geq I_3$

(C) $I_1 > I_3 > I_4$ (D) $I_1 \geq I_3 \geq I_4$

Q11. which of the following is/are not Incorrect?

(A) $\frac{\pi}{6} < \int_0^1 \frac{dx}{\sqrt{4-x^2-x^3}}\,dx < \frac{\pi\sqrt{2}}{8}$ (B) $2e^{-1/4} < \int_0^2 e^{x^2-x}\,dx < 2e^2$

(C) $\frac{1}{3} < \int_0^1 x^{(\sin x+\cos x)^2}\,dx < \frac{1}{2}$.

(D) $\frac{1}{2} \leq \int_0^2 \frac{dx}{2+x^2} \leq \frac{5}{6}$

Q12. solve $\dfrac{\int_0^{\pi/2} (\sin x)^{\sqrt{2}+1}\,dx}{\int_0^{\pi/2} (\sin x)^{\sqrt{2}-1}\,dx}$

Q13. $\lim_{n\to\infty} \left[\sum_{k=1}^{n} \left(\frac{1}{{}^nC_k}\right)\right]^n$

Q14. Let $u = \int_0^\infty \frac{dx}{x^4+7x^2+1}$ and $v = \int_0^\infty \frac{x^2\,dx}{x^4+7x^2+1}$ then

(A) $v > u$ (B) $6v = \pi$ (C) $3u + 2v = \frac{5\pi}{6}$ (D) $u + v = \frac{\pi}{3}$.

Q15. Let $L = \lim_{y \to 0} \frac{1}{y} \int_0^{\pi/2} \tan(y \sin x)\,dx$ and

$l = \lim_{k \to \infty} \int_0^\infty \left(1 - \frac{x}{k}\right)^k e^{x/3}\,dx$ then

(A) $3l = 4L$ (B) $3L = 4l$ (C) $L + 2l = 5$ (D) $2(l+L) = 7$.

Q16. The value of $\frac{\int_0^1 (1-x^2)^{1/3}\,dx}{\int_0^1 (1-x^3)^{1/2}\,dx}$ is equal to

(A) 1 (B) 2 (C) $\frac{1}{2}$ (D) $\frac{1}{6}$.

Q17. ~~The value of $\int_0^1 (1+x^2)^{1/3}\,dx$~~ Let the functions $f(x)$, $g(x)$ be differentiable and strictly increasing $\forall\, x \in R$. Also $f(0) = 2$, $f(1) = 3$ & $f'(x)$ is continuous. If $f^2(x) = 1 + g^2(x)$ & $J = \int_0^1 \left(\frac{g(x)}{f(x)}\right)' \frac{1}{g(x)}\,dx$ then the value of $6J + 1$ is

(A) $\ln\left(\frac{3}{2}\right)$ (B) $\ln\left(\frac{27}{8}\right)$ (C) $\ln\left(\frac{9}{4}\right)$ (D) NOT.

Q18. The value of the definite integral $\int_0^\pi \left((1 - x\sin 2x)\left(e^{\cos^2 x} + (1 + x\sin 2x)\,e^{\sin^2 x}\right)\right)dx$ is equal to

(A) π (B) πe (C) $\pi(e-1)$ (D) $\pi(e+1)$.

Q19. which of the following statements is/are true?

(A) Let f be strictly increasing continuous function defined on the set of real numbers. If $f(1) = 2$, $f(2) = 4$ then

$\int_1^2 f(x)\,dx$ can be equal to $\frac{25}{8}$.

(B) Let $f: R^+ \to R^+$ is such that $f''(x) > 0$, $\forall\, x \in [a,b]$, then the value of $\int_a^b f(x)\,dx$ can not exceed $\frac{(f(a)+f(b))(b-a)}{2}$

(C) the value of $\int_0^1 \frac{dx}{1+x^{\pi/2}} < \frac{\pi}{4}$

(D) the value of $\int_0^1 \frac{dx}{1+x^{\pi/2}} > \ln 2$.

Q20. If $I = \int_0^{\pi/2} e^{-d\sin x}\,dx$, $d \in (0,\infty)$ then which of the following is/are true?

(A) $I < \frac{\pi}{2}$ (B) $I > \frac{\pi}{2}(e^{-d}+1)$

(C) $I > \frac{\pi}{2}e^{-d}$ (D) $I > 0$.

Q21. If $a, b, k \in R$, $a < b$ and that f is continuous on $[a,b]$ and ~~the~~ if the minimum value of $\int_a^b (f(x)-k)^2\,dx$ equals $d\int_a^b (f(x))^2\,dx + \frac{u}{b-a}\left(\int_a^b f(x)\,dx\right)^2$ then the value of $d+u$ equals

Q22. The value of $\int_0^2 \left(\sqrt{1+x^3} + \sqrt[3]{x^2+2x}\right)dx$ equals

Q23. If the integral $\int_0^1 \frac{1}{\sqrt{1+x}+\sqrt{1-x}+2}\,dx = p\sqrt{2} + q + \frac{r\pi}{2}$ where $p, q, r \in z$

(A) $p = 2$ (B) $q = 1$ (C) $r = -1$ (D) $|p| = 2$.

Q24. If $\int_0^1 \frac{\tan^{-1}(x\sqrt{x^2+2})}{\sqrt{x^2+2}} \cdot \frac{dx}{x^2+1} = \frac{\pi^2}{k}$ then

(A) $p = 2$ (B) $q = 1$ (C) $r = -1$ (D) $|p| = 2$

Q25. If $\int_0^1 \frac{\tan^{-1}(x\sqrt{x^2+2})}{\sqrt{x^2+2}} \cdot \frac{dx}{x^2+1} = \frac{\pi^2}{k}$ then which is/are correct?

(A) k is a perfect square

(B) sum of digits of k is 9

(C) number of positive integral divisors of k is 5

(D) $k > 11$.

Q4. $nC_0\, nC_1\, nC_2 \cdots nC_n = \left(\frac{n!}{0!\,n!}\right)\left(\frac{n!}{1!\,(n-1)!}\cdots\frac{n!}{n!\,0!}\right)$

$$= \frac{(n!)^{n+1}}{(1!\,2!\,3!\cdots n!)^2}$$

$$(nC_0\, nC_1\, nC_2 \cdots nC_n)^{1/n(n+1)} = \left[\frac{\frac{(n!)^{n+1}}{n^{n(n+1)}}}{\frac{(1!\,2!\,3!\cdots n!)^2}{n^{n(n+1)}}}\right]^{\frac{1}{n(n+1)}}$$

$$= \frac{p}{q}$$

where $p = \left[\frac{(n!)^{n+1}}{n^{n(n+1)}}\right]^{\frac{1}{n(n+1)}}$, $q = \left[\frac{(1!\,2!\,3!\cdots n!)^2}{n^{n(n+1)}}\right]^{\frac{1}{n(n+1)}}$

$p = \left(\frac{n!}{n^n}\right)^{1/n}$, $\log p = \frac{1}{n}\sum_{r=1}^{n} \log\frac{r}{n}$

$= \int_0^1 \log x\, dx = |x\log x - x|_0^1 = -1$

$q = \left[\frac{1!\,2!\,3!\,4!\,5!\cdots n!}{n^{\frac{n(n+1)}{2}}}\right]^{2/n(n+1)}$

$\log q = \frac{2}{n(n+1)} \log\left(\frac{0!\,1!\,2!\,3!\cdots n!}{n^{\frac{n(n+1)}{2}}}\right)$

$= \frac{2}{n(n+1)} \sum_{k=0}^{n} \log\left(\frac{k!}{n^k}\right)$

$= \frac{2}{n+1}\left[\sum_{k=0}^{n}\left[\frac{1}{n}\sum_{r=1}^{k}\log\left(\frac{r}{n}\right)\right]\right.$

$= \frac{2}{n+1} \sum_{k=0}^{n} \int_0^{k/n} \log t \, dt = 2 \int_0^1 \left(\int_0^x \log t \, dt \right) dx$

$= 2 \int_0^1 (x \log x - x) \, dx = 2 \left| \frac{x^2}{2} \log x - \frac{x^2}{4} - \frac{x^2}{2} \right|_0^1$

$= -2\left(\frac{3}{4}\right) = -\frac{3}{2}$, $\frac{p}{q} = \sqrt{4}$.

Q5. $\lim_{n \to \infty} \prod_{r=1}^{n} \frac{4r^2}{4r^2 - 1}$

$$\lim_{n \to \infty} \frac{2^{2n} (n!)^2}{(3 \cdot 5 \cdot 7 \cdot 9 \cdots (2n-1))^2 (2n+1)}$$

$$= \lim_{n \to \infty} \left(\frac{2^n n!}{3 \cdot 5 \cdot 7 \cdots (2n-1)} \right)^2 \times \frac{1}{2n+1}$$

$$= \lim_{n \to \infty} \frac{(2^n n! \times 2^n n!)^2}{(2n!)^2} \times \frac{1}{2n+1}$$

$$= \lim_{n \to \infty} \left[\frac{2^{2n} (n!)^2}{(2n)!} \right]^2 \times \frac{1}{2n+1}$$

Since $\lim_{n \to \infty} n! \sim \sqrt{2\pi n} \left(\frac{n}{e}\right)^n$

$$= \lim_{n \to \infty} \frac{\left[2^{2n} \left[\sqrt{2\pi n} \left(\frac{n}{e}\right)^n \right]^2 \times \frac{1}{2n+1} \right]}{\sqrt{4\pi n} \left(\frac{2n}{e}\right)^{2n}}$$

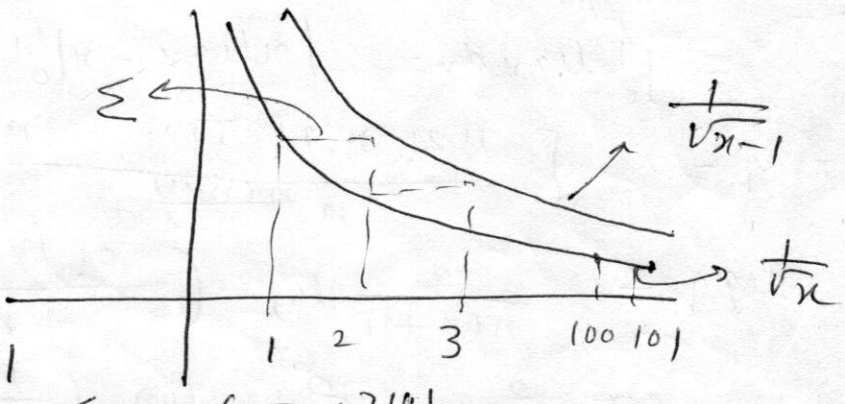
$$= \lim_{n \to \infty} \frac{\left(2^{2n} (2\pi n) \left(\frac{n}{e}\right)^{2n} \right)}{2^{2n} \left(\frac{n}{e}\right)^{2n} \sqrt{4\pi n}} \times \frac{1}{2n+1}$$

$$= \lim_{n \to \infty} \frac{\pi n}{2n+1} = \frac{\pi}{2}, \quad [37 \times p] = \left[37 \times \frac{\pi}{2}\right] = 58.$$

Q6 $g(x) = \frac{1}{\sqrt{x}}$

$\sum_{x=1}^{100} g(x) > \int_1^{100} \frac{1}{\sqrt{x}}$

$> 2 [\sqrt{x}]_1^{100} > 18$



$\sum_{x=1}^{100} g(x) < \int_1^{101} \frac{1}{\sqrt{x-1}} dx + 1$

$< 2 (\sqrt{x-1})]_1^{101} + 1 \leq 19$

$\therefore \left[\sum_{k=1}^{100} g(x) \right] = 18 \Rightarrow \frac{k}{2} = 9.$

Q7. $S = \int_1^{e^2} \frac{\ln x}{\sqrt{x}}\, dx$

$f(x) = \frac{\ln x}{\sqrt{x}}$, $f'(x) = \frac{\frac{\sqrt{x}}{x} - \ln x \times \frac{1}{2\sqrt{x}}}{x}$

$= \frac{2 - \ln x}{2x^{3/2}} > 0 \Rightarrow \ln x < 2 \Rightarrow x < e^2$

(A) $S < \int_1^{e^2} \frac{\ln e^2}{\sqrt{e^2}}\, dx < \frac{2}{e} \int_1^{e^2} dx < 2\left(e - \frac{1}{e}\right)$

(C) Let $\ln x = t \Rightarrow dx = e^t\, dt$

$S = \int_0^2 \frac{t\, e^t\, dt}{\sqrt{e^t}} = \int_0^2 t \sqrt{e^t}\, dt$

(D) Let $\sqrt{x} = t$, $S = \int_1^e \frac{\ln t^2}{t} \times 2t\, dt = 4\int_1^e \ln t\, dt$

$= 4(t \ln t - t)\Big]_1^e = 4 \Rightarrow 3 < S < 5$

(B) $e - 1 < \left(2 + \frac{1}{\sqrt{e}}\right) = 4.4787$.

$S < (e-1)\left(2 + \frac{1}{\sqrt{e}}\right)$

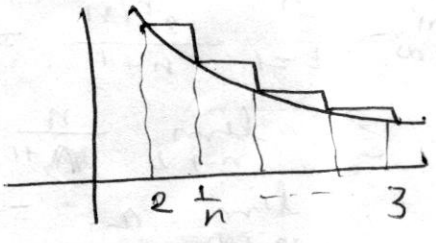
Q8. $T(n) = \sum_{r=2n}^{3n-1} \frac{r}{n^2 + r^2} = \sum_{r=2n}^{3n-1} \frac{1}{n}\left(\frac{\frac{r}{n}}{1 + \left(\frac{r}{n}\right)^2}\right)$

$S(n) = \sum_{r=2n+1}^{3n} \frac{r}{r^2 + n^2} = \sum_{r=2n+1}^{3n} \frac{1}{n}\left(\frac{r/n}{1 + \frac{r^2}{n^2}}\right)$

$f(x) = \frac{x}{1+x^2}$

$f'(x) = \frac{(1+x^2) \times 1 - x \times 2x}{(1+x^2)^2} = \frac{1 - x^2}{(1+x^2)^2}$

$f'(x) < 0$ for $x > 1$

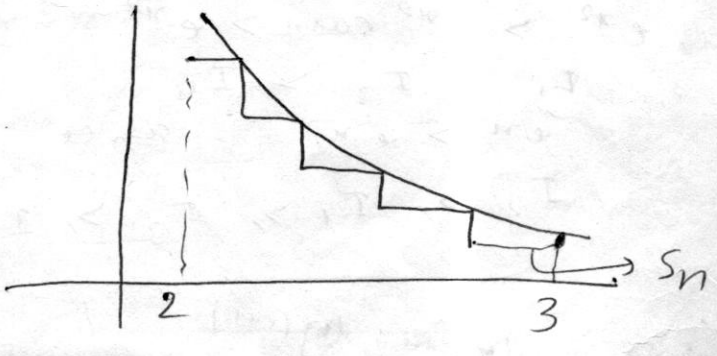


$\Rightarrow T(n) > \int_2^3 f(x)\, dx$

$\Rightarrow S(n) < \int_2^3 f(x)\, dx$

$\int_2^3 f(x)\, dx = \int_2^3 \frac{x}{1+x^2}\, dx = \left| \frac{\ln(1+x^2)}{2} \right|_2^3 = \frac{\ln 2}{2}$

$S_n < \frac{\ln 2}{2} < T_n$



Q9. (A) $\int_0^\infty \frac{\sin t \cos t}{t}\,dt = \frac{\pi}{4}$

$\int_0^\infty \frac{\sin 2t}{2t}\,dt = \frac{\pi}{4}$, put $2t = x$

$= \int_0^\infty \frac{\sin u}{u}\,\frac{du}{2} = \frac{\pi}{4} \Rightarrow \int_0^\infty \frac{\sin u}{u}\,du = \frac{\pi}{2}$

(B) $\int_1^k \ln x\,dx > \frac{1}{2}\ln k + \ln((k-1)!)$

$\ln x$ is a concave function.

then $\int_i^{i+1} \ln x\,dx > \frac{1}{2}(\ln(i) + \ln(i+1)) - 1$

$\Rightarrow \sum_{i=1}^{k-1} \int_i^{i+1} \ln x\,dx > \sum_{i=1}^{k-1} \frac{1}{2}(\ln(i) + \ln(i+1))$

$\Rightarrow \int_1^k \ln x\,dx > \frac{\ln k}{2} + \ln((k-1)!)$.

(C) $\int_1^k \ln x\,dx = (x\ln x - x)\Big]_1^k = k\ln k - k + 1$

$\Rightarrow k\ln k - k + 1 > \frac{1}{2}\ln k + \ln((k-1)!)$

$\Rightarrow k\ln\left(\frac{k}{e}\right) + 1 > \ln(\sqrt{k}\,(k-1)!)$

$\Rightarrow \ln\left(e^1\left(\frac{k}{e}\right)^k\right) > \ln(\sqrt{k}\,(k-1)!)$

$\Rightarrow e\left(\frac{k}{e}\right)^k > \sqrt{k}\,(k-1)!$, $k! < \sqrt{k}\,e\left(\frac{k}{e}\right)^k$

(D) $a_n = \sum_{k=1}^{n} \frac{2^{k/n}}{n+\frac{1}{k}}$, $\sum_{k=1}^{n} \frac{2^{k/n}}{n+1} < a_n < \sum_{k=1}^{n} \frac{2^{k/n}}{n}$

$\lim_{n\to\infty} a_n < \lim_{n\to\infty} \sum_{k=1}^{n} \frac{2^{k/n}}{n} \le \int_0^1 2^x\,dx$

$\le \frac{|2^x|_0^1|}{\ln 2} \le \frac{1}{\ln 2}$

$\lim_{n\to\infty} \sum_{k=1}^{n} \frac{2^{k/n}}{n+1} = \lim_{n\to\infty} \frac{n}{n+1} \sum_{k=1}^{n} \frac{2^{k/n}}{n}$

$\le \lim_{n\to\infty} \frac{n}{n+1} \int_0^1 2^x\,dx = \frac{1}{\ln 2}$

$\therefore \lim_{n\to\infty} a_n = \frac{1}{\ln 2}$

Q10. $I_1 = \int_0^{\pi/4} e^{x^2}\,dx$, $I_2 = \int_0^{\pi/4} e^x\,dx$

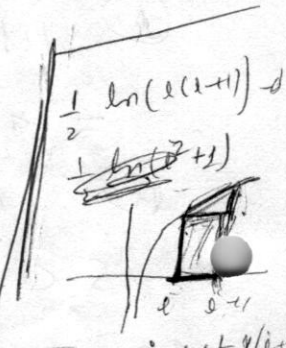
$I_3 = \int_0^{\pi/4} e^{x^2}\cos x\,dx$, $I_4 = \int_0^{\pi/4} e^{x^2}\sin x\,dx$.

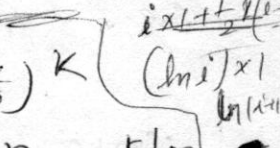
$e^{x^2} > e^{x^2}\cos x > e^{x^2}\sin x$ $\because \cos x > \sin x$ $\left(0 < \frac{\pi}{4}\right)$

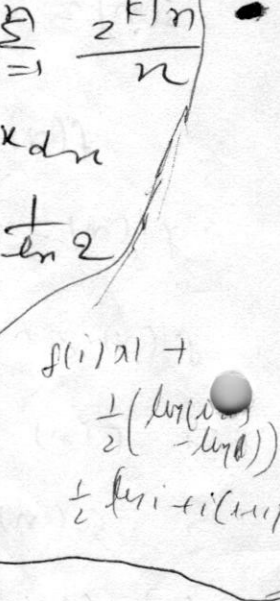
$I_1 > I_3 > I_4$

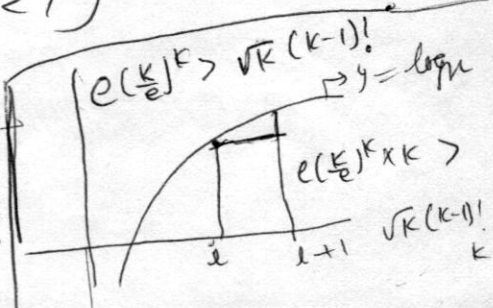
$e^x > e^{x^2}$ since $x < 1$ $\left(\frac{\pi}{4} < 1\right)$

$I_2 > I_1 > I_3 > I_4$









Q11. (A) $\int_0^1 \frac{dx}{\sqrt{4-x^2-x^3}}$

$$\int_0^1 \frac{dx}{\sqrt{4-x^2}} < \int_0^1 \frac{dx}{\sqrt{4-x^2-x^3}} < \int_0^1 \frac{dx}{\sqrt{4-2x^2}}$$

$$\left|\sin^{-1}\frac{x}{2}\right|_0^1 < \int_0^1 \frac{dx}{\sqrt{4-x^2-x^3}} < \int_0^1 \frac{1}{\sqrt{2}}\sin^{-1}\left(\frac{x}{\sqrt{2}}\right)\Big]_0^1$$

$$\pi/6 \le I \le \frac{\pi\sqrt{2}}{8}.$$

(B) $2e^{-1/4} < \int_0^2 e^{x^2-x}\,dx < 2e^2$

min value of x^2-x in $[0,2] = -\frac{1}{4}$ at $x=\frac{1}{2}$

& max $= 2$ at $x=2$

$$\int_0^2 e^{-1/4}\,dx < \int_0^2 e^{x^2-x}\,dx < \int_0^2 e^2\,dx$$

(C) $\frac{1}{3} < \int_0^1 x^{(\sin x+\cos x)^2}\,dx$, $(\sin x+\cos x)^2 = 1+\sin 2x$

$1 < (\sin x+\cos x)^2 < 2$, $x^1 > x^{(\sin x+\cos x)^2} > x^2$.

since $0<x<1$

$\int_0^1 x\,dx > \int_0^1 x^{(\sin x+\cos x)^2}\,dx > \int_0^1 x^2\,dx$

(D) $\int_0^2 \frac{dx}{2+x^2} = \frac{1}{\sqrt{2}}\left(\tan^{-1}\frac{x}{\sqrt{2}}\right)_0^2 = \frac{\tan^{-1}\sqrt{2}}{\sqrt{2}}$

Q12. $$I = \frac{\int_0^{\pi/2} (\sin x)^{\sqrt{2}+1}\,dx}{\int_0^{\pi/2} (\sin x)^{\sqrt{2}-1}\,dx}$$

$$I_1 = \int_0^{\pi/2} (\sin x)^{\sqrt{2}+1}\,dx$$

$$= \int_0^{\pi/2} (\sin x)^{\sqrt{2}} \sin x\,dx$$

$$= \left|(\sin x)^{\sqrt{2}}(-\cos x)\right|_0^{\pi/2} - \int_0^{\pi/2} \sqrt{2}\,(\sin x)^{\sqrt{2}-1}\cos x(-\cos x)\,dx$$

$$I = \int_0^{\pi/2} \sqrt{2}\,(\sin x)^{\sqrt{2}-1}\cos 2x\,dx$$

$$= \int_0^{\pi/2} \sqrt{2}\,(\sin x)^{\sqrt{2}-1}(1-\sin^2 x)\,dx$$

$$I = \sqrt{2}\int_0^{\pi/2} (\sin x)^{\sqrt{2}-1}\,dx - \sqrt{2}\,I$$

$$I = \frac{\sqrt{2}}{1+\sqrt{2}}\int_0^{\pi/2} (\sin x)^{\sqrt{2}-1}\,dx$$

$$\therefore \frac{\int_0^{\pi/2} (\sin x)^{\sqrt{2}+1}\,dx}{\int_0^{\pi/2} (\sin x)^{\sqrt{2}-1}\,dx} = \frac{\sqrt{2}}{1+\sqrt{2}}$$

Q13. $\lim_{n\to\infty} \left[\sum_{k=1}^{n} \left(\frac{1}{nCk} \right) \right]^{n}$

$= \left[\lim_{n\to\infty} \left(\sum_{k=1}^{n} \frac{k!\,(n-k)!}{n!} \right)^{\frac{1}{n^2}} \right]^{n^3}$

$y = \lim_{n\to\infty} \sum_{k=1}^{n} \left(\frac{k!\,(n-k)!}{n^n} \right)^{\frac{1}{n^2}}$

$\log y = \lim_{n\to\infty} \frac{1}{n^2} \sum_{k=1}^{n} \log\left(\frac{k!}{n^k} \right) + \log\left(\frac{(n-k)!}{n^{n-k}} \right)$

$= \frac{1}{n} \sum_{k=1}^{n} \left[\int_0^{k/n} \log t + \log(1-t)\, dt \right]$

$= \int_0^1 \left[\int_0^x \log t + \log(1-t)\, dt \right] dx$

$\log y = \int_0^1 \left[t\log t - t + t\log(1-t) + \int \frac{t}{1-t}\, dt \right]_0^x dx$

$= \int_0^1 \left[t\log t - t + t\log(1-t) - t - \log(1-t) \right]_0^x dx$

$= \int_0^1 \left[t\log t - t + t\log(1-t) - t - \log(1-t) \right]_0^x dx$

$= \int_0^1 \left(x\log x - 2x + x\log(1-x) - \log(1-x) \right) dx$

$= \int_0^1 x\log x\, dx - \int_0^1 2x\, dx - \int_0^1 (1-x)\log(1-x)\, dx$

$= \int_0^1 x\log x\, dx - \int_0^1 2x\, dx - \int_0^1 x\log x\, dx$

$= -\int_0^1 2x\, dx = -1, \quad y = e^{-1}$

$\lim_{n\to\infty} \left[\sum_{k=1}^{n} \left[\frac{k!\,(n-k)!}{n^n} \right] \right]^{\frac{1}{n^2}} = e^{-1}$

Let $y = \lim_{n\to\infty} \left(\frac{n!}{n^n} \right)^{\frac{1}{n}}$

$\log y = \lim_{n\to\infty} \frac{1}{n} \log\left(\frac{n!}{n^n} \right)$

$= \int_0^1 \log x\, dx = (x\log x - x)\Big]_0^1 = -1$

$y = e^{-1}$

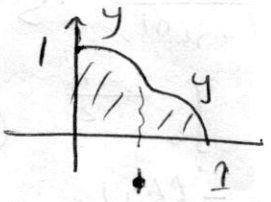
$\lim_{n\to\infty} \left(\frac{n!}{n^n} \right)^{\frac{1}{n}} = e^{-1}$

$\therefore \lim_{n\to\infty} \left[\sum_{k=1}^{n} \frac{(k!)\,(n-k)!}{n!} \right]^{n}$

$= \lim_{n\to\infty} \left[\frac{\sum_{k=1}^{n} (k!)\,(n-k)!^{1/n^2}}{\left(\frac{n!}{n^n} \right)^{1/n^2}} \right]$

$= \lim_{n\to\infty} \left(\frac{e^{-1}}{e^{-1/n}} \right)^{n^3} = 0$

Q14. $\dfrac{\int_0^1 (1-x^2)^{1/3}\,dx}{\int_0^1 (1-x^3)^{1/2}\,dx}$, $\int_0^1 (1-x^2)^{1/3}\,dx$



Let $y = (1-x^2)^{1/3}$

$= \int_0^1 y\,dx =$ Area under curve

$= \int_0^1 x\,dy = \int_0^1 \sqrt{1-y^2}\,dy$

$y^3 = 1-x^2$, $x = \sqrt{1-y^3}$, $\int_0^1 \sqrt{1-x^3}\,dx$

$\dfrac{\int_0^1 (1-x^3)^{1/3}\,dx}{\int_0^1 \sqrt{1-x^3}\,dx} = 1$

Q15. $J = \int_0^1 \left(\dfrac{g(x)}{f(x)}\right)' \dfrac{dx}{g(x)}$, $f'(x) = 1 + g(x)$

$J = \int_0^1 \dfrac{(g'(x)f(x) - f'(x)g(x))}{f'(x)} \dfrac{dx}{g(x)}$

$= \int_0^1 \dfrac{g'(x)}{f(x)g(x)}\,dx - \int_0^1 \dfrac{f'(x)}{f'(x)}\,dx$

$= 2f(x)f'(x) = 2g(x)g'(x)$

$= \int_0^1 \dfrac{f'(x)}{g'(x)}\,dx - \int_0^1 \dfrac{f'(x)}{f'(x)}\,dx$

$= \int_0^1 \dfrac{f'(x)}{f(x)-1}\,dx - \int_0^1 \dfrac{f'(x)}{f(x)}\,dx$

$= 1 \cdot \dfrac{1}{2} \ln\left(\dfrac{f(x)-1}{f(x)+1}\right) + \dfrac{1}{f(x)}\Big|_0^1$

$= \dfrac{1}{2}\ln\dfrac{1}{2} + \dfrac{1}{3} - \left(\dfrac{1}{2}\ln\left(\dfrac{1}{3}\right) + \dfrac{1}{2}\right)$

$= \dfrac{1}{2}\ln\dfrac{3}{2} - \dfrac{1}{6}$, $6J + 1 = 3\ln\left(\dfrac{3}{2}\right) = \ln\left(\dfrac{27}{8}\right)$

Q16. $\int_0^{\pi} (1 - x\sin 2x)\left(e^{\cos^2 x} + (1 + x\sin 2x)e^{\sin^2 x}\right)dx$

$= \int_0^{\pi} e^{\cos^2 x}(1 - x\sin 2x) + (1 + x\sin 2x)e^{\sin^2 x}\,dx$

$= \int e^{\cos 2x}\,dx = x e^{\cos 2x} - \int x(-\sin 2x)e^{\cos 2x}\,dx$

$x e^{\cos 2x} = \int (e^{\cos 2x} - x\sin 2x\, e^{\cos 2x})\,dx$

$\int e^{\sin^2 x}\,dx = x e^{\sin^2 x} - \int x\sin 2x\, e^{\sin^2 x}$

$x e^{\sin^2 x} = \int (1 + x\sin 2x)e^{\sin^2 x}\,dx$

$= 1 \cdot x e^{\cos 2x} + x e^{\sin^2 x}\Big|_0^{\pi}$

$= \pi(e+1)$

Q17. (C) $\int_0^1 \frac{dx}{1+x^{11/2}} < \int_0^1 \frac{dx}{1+x^2} < |\tan^{-1}x|_0^1 < \frac{\pi}{4}$

(D) $\int_0^1 \frac{dx}{1+x^{11/2}} > \int_0^1 \frac{dx}{1+x} > |\ln(1+x)|_0^1 > \ln 2$

(B) $I = \int_0^{\frac{b-a}{2}} (f(a+x) - f(b-x))\,dx$

$\frac{f(a+x)-f(a)}{x} = f'(c_1)$, $a < c_1 < a+x$

$\frac{f(b)-f(b-x)}{x} = f'(c_2)$, $b-x < c_2 < b$

$f'(c_2) \geq f'(c_1)$, $\frac{f(b)-f(b-x)}{x} \geq \frac{f(a+x)-f(a)}{x}$

$f(a)+f(b) \geq f(a+x)+f(b-x)$

$I \leq f(a)+f(b)\int_0^{\frac{b-a}{2}} dx \leq \frac{b-a}{2}(f(a)+f(b))$

(A) As from part B integral should be less than $\left[\frac{f(1)+f(2)}{2}\right](2-1)$ Hence not possible.

Q18. $I = \int_0^{\pi/2} e^{x\sin x}\,dx$, $1 > e^{x\sin x} > e^{-x}$

$I > \int_0^{\pi/2} e^{-x}\,dx = e^{-x}\Big|_0^{\pi/2}$

$I < \int_0^{\pi/2} e^0\,dx < \frac{\pi}{2}$, $I > \frac{\pi}{2}e^{-2} \Rightarrow I > 0$

$I < \pi/2 \rightarrow$ (A) is wrong.

Q19. $\int_a^b [f(x)-k]^2\,dx = g(k)$

$g'(k) = \int_a^b 2(f(x)-k)(-1)\,dx$

$= 2k(b-a) - 1\int_a^b f(x)\,dx$

$g(k)$ is min at $k = \frac{\int_a^b f(x)\,dx}{b-a}$

$\int_a^b [f(x)-k]^2\,dx = \int_a^b f^2(x)\,dx + k^2(b-a) - 2k\int_a^b f(x)\,dx$

$= \int_a^b f^2(x)\,dx + \frac{[\int_a^b f(x)\,dx]^2}{b-a} - 2\frac{[\int_a^b f(x)\,dx]^2}{b-a}$

$= \int_a^b f^2(x)\,dx - \frac{[\int_a^b f(x)\,dx]^2}{b-a} \Rightarrow$ Ans $= 0$.

Q20. $\int_0^2 \left(\sqrt{1+x^3} + \sqrt[3]{x^2+2x}\right)dx$

Let $y = \sqrt{1+x^3}$, $y^2 = 1+x^3$, $x = (y^2-1)^{1/3}$

$\Rightarrow \int_0^2 \sqrt{1+x^3}\,dx = \int_1^3 y\frac{d(y^2-1)^{1/3}}{dy}\,dy$.

$= |y(y^2-1)^{1/3}|_1^2 - \int_1^3 (y^2-1)^{1/3}\,dy$

$= 3(3^2-1)^{1/3} - 1\cdot(1^2-1)^{1/3} - \int_0^2 [(t+1)^2-1]^{1/3}\,dt$

$\Rightarrow \int_0^2 \sqrt{1+x^3}\,dx = 6 - \int_0^2 (t^2+2t)^{1/3}\,dt$ $\quad y = t^2+1$

$= \int_0^2 \sqrt{1+x^3}\,dx + \int_0^2 (t^2+2t)^{1/3}\,dt = 6$

Ans 6

Q21. $\int_0^1 \frac{dx}{\sqrt{1+x}+\sqrt{1-x}+2}$,

let $x = \sin\theta$, $\sqrt{1+x} = \cos\frac{\theta}{2} + \sin\frac{\theta}{2}$

$\sqrt{1-x} = \cos\frac{\theta}{2} - \sin\frac{\theta}{2}$

$\int_0^{\pi/2} \frac{\cos\theta\,d\theta}{2\cos\frac{\theta}{2}+2} = \int_0^{\pi/2} \frac{2\cos^2\frac{\theta}{2}-1}{2\cos\frac{\theta}{2}+2}\,d\theta$

$= \int_0^{\pi/2} \left(\frac{\cos^2\theta/2 - \frac{1}{2}}{\cos\theta/2 - 1}\right) d\theta = \int_0^{\pi/2} \left(\cos\frac{\theta}{2} - \frac{1}{2}\right) + \frac{1}{2}\left[\frac{1}{\cos\frac{\theta}{2}+1}\right] d\theta$

$= 2\sqrt{2} - \frac{\pi}{2} + 1$

Q22. $\int_0^1 \frac{\tan^{-1}\left(x\sqrt{x^2+2}\right)}{\sqrt{x^2+1}} \frac{dx}{x^2+1}$

let $x^2+1 = \sec\theta$, $2x\,dx = \sec\theta\tan\theta\,d\theta$

then $x\sqrt{x^2+2} = \sqrt{x^4+2x^2} = \sqrt{(x^2+1)^2-1}$

$= \sqrt{\sec^2\theta - 1} = \tan\theta$

$\int_0^{\pi/4} \frac{\tan^{-1}(\tan\theta)}{(x\sqrt{x^2+2})} \frac{x\,dx}{\sec\theta} = \int_0^{\pi/4} \frac{\theta}{2} \frac{\sec\theta\tan\theta\,d\theta}{\tan\theta\sec\theta}$

$= \int_0^{\pi/2} \theta\,d\theta = \pi^2/4$, $k = 64 = 26$.

Q14. $u = \int_0^\infty \frac{dx}{x^4+7x^2+1}$, $v = \int_0^\infty \frac{x^2\,dx}{x^4+7x^2+1}$

$u+v = \int_0^\infty \frac{(x^2+1)\,dx}{x^4+7x^2+1} = \int_0^\infty \frac{\left(1+\frac{1}{x^2}\right)dx}{x^2+\frac{1}{x^2}+7} = \int_0^\infty \frac{\left(1+\frac{1}{x^2}\right)dx}{\left(x-\frac{1}{x}\right)^2+9}$

let $\alpha = x - \frac{1}{x}$, $d\alpha = \left(1+\frac{1}{x^2}\right)dx$

$u+v = \int_{-\infty}^{\infty} \frac{d\alpha}{\alpha^2+9} = \frac{1}{3}\tan^{-1}\left(\frac{\alpha}{3}\right)\Big|_{-\infty}^{\infty} = \frac{\pi}{3}$

$v-u = \int_{-\infty}^{\infty} \frac{x^2-1}{x^4+7x^2+1}\,dx = \int_{-\infty}^{\infty} \frac{d\left(x+\frac{1}{x}\right)}{\left(x+\frac{1}{x}\right)^2+5} = 0$

$\Rightarrow v=u$, $v+u=\frac{\pi}{3} \Rightarrow u=\frac{\pi}{6}$, $v=\frac{\pi}{6}$

Q15. $L=\lim_{y\to 0}\int_0^{\pi}\frac{\tan(y\sin x)\,dx}{y}$

Using L'Hospital

$$L=\lim_{y\to 0}\int_0^{\pi/2}\frac{\sec^2(y\sin x)\sin x\,dx}{1}$$

$$=\int_0^{\pi/2}\sin x\,dx=-\cos x\Big|_0^{\pi}=2$$

$$l=\lim_{k\to\infty}\int_0^{\infty}\left(1-\frac{x}{k}\right)^k e^{x/3}\,dx$$

$$=\int_0^{\infty}e^{-x}e^{x/3}\,dx=\int_0^{\infty}e^{-2x/3}\,dx$$

$$=-\frac{3}{2}e^{-2x/3}\Big|_0^{\infty}=-\frac{3}{2}\times -1=\frac{3}{2}$$

$$\int_0^{\infty}e^{-x}x^{n-1}$$

$$\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^x$$

$$\lim_{x\to\infty}(1+x)^{1/x}$$

$$(1+\infty)^{\infty}=e$$

$$A = \int_0^{\pi/2} \frac{\sin x}{x}\,dx$$

$$\sin x < x < \tan x$$

$$\Rightarrow \frac{\sin x}{x} < 1$$

$$\frac{\sin x}{x} > \cos x$$

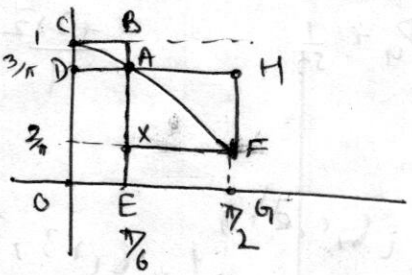
$$\Rightarrow \int_0^{\pi/2} 1\,dx > A > \int_0^{\pi/2} \cos x\,dx$$

$$\Rightarrow 1 < A < \pi/2$$

$$A = \int_0^{\pi/6} \frac{\sin x}{x}\,dx + \int_{\pi/6}^{\pi/2} dx$$

let $f(x) = \frac{\sin x}{x}$

~~$f(x)$~~ = graph of $f(x)$ is



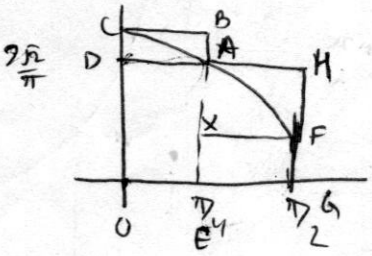
$$\int_0^{\pi/2} f(x)\,dx \leq \text{area}(OEBC) + \text{area}(EGHA)$$

$$= \pi/6 + \frac{3}{\pi} \times \pi/3 = 1 + \pi/6$$

$$\int_0^{\pi/2} f(x)\,dx \geq \text{area}(OEAD) + \text{area}(EGFX)$$

$$= \pi/6 \times 3/\pi + \frac{2}{\pi} \times \pi/3 = \frac{2}{3} + 1/2 = 7/6$$

Similarly at $x = \pi/4$



$$\int_0^{\pi/2} f(x)\,dx \geq \text{area}(OEAD) + \text{area}(EGFX)$$

$$= \frac{2\sqrt{2}}{\pi} \times \pi/4 + \frac{2}{\pi} \times \pi/4$$

$$= \frac{1}{\sqrt{2}} + \frac{1}{2} = \frac{\sqrt{2}+1}{2}$$

$$\int_0^{\pi/2} f(x)\,dx \leq \text{area}(OEBC) + \text{area}(EGHA)$$

$$= 1 \times \pi/4 + \frac{2\sqrt{2}}{\pi} \times \pi/4$$

$$= \pi/4 + \frac{1}{\sqrt{2}} = \frac{\pi + 2\sqrt{2}}{4}$$

1, 4, 5, 9 (b, c, d),
11, 13, 16, 18, 23, 20, 21

Since $\frac{\sin x}{x}$ is a decreasing function

$$\Rightarrow \frac{\sin(\pi/2)}{\pi/2} \leq \frac{\sin x}{x} \leq \frac{\sin(0)}{0}$$

$$\Rightarrow \frac{2}{\pi} \leq \frac{\sin x}{x} \leq 1$$

$$\Rightarrow 1 \leq \int_0^{\pi/2} \frac{\sin x}{x}\,dx \leq \pi/2$$

$$A = \int_0^{\pi/6} \frac{\sin x}{x}\,dx + \int_{\pi/6}^{\pi/2} \frac{\sin x}{x}\,dx$$

for $x \in [0, \pi/6]$

$$\Rightarrow \frac{3}{\pi} \leq \frac{\sin x}{x} \leq 1$$

for $x \in [\pi/6, \pi/2]$

$$\frac{2}{\pi} \leq \frac{\sin x}{x} \leq \frac{3}{\pi}$$

$$\Rightarrow \frac{1}{2} \leq \int_0^{\pi/6} \frac{\sin x}{x}\,dx \leq \pi/6$$

$$\frac{2}{3} \leq \int_{\pi/6}^{\pi/2} \frac{\sin x}{x}\,dx \leq 1$$

Adding

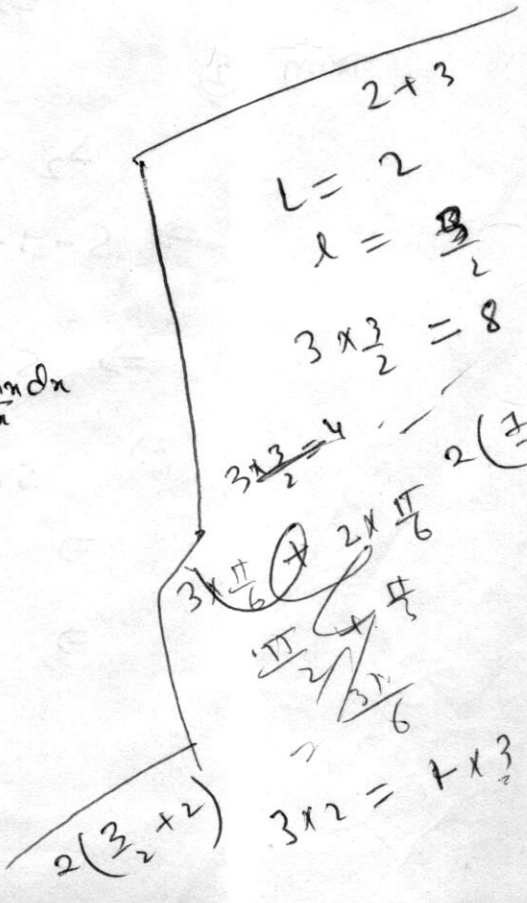
$$\Rightarrow \frac{7}{6} \leq \int_0^{\pi/2} \frac{\sin x}{x} \leq \frac{\pi+6}{6}$$

Similarly $\int_0^{\pi/2} \frac{\sin x}{x}\,dx = \int_0^{\pi/4} \frac{\sin x}{x}\,dx + \int_{\pi/4}^{\pi/2} \frac{\sin x}{x}\,dx$

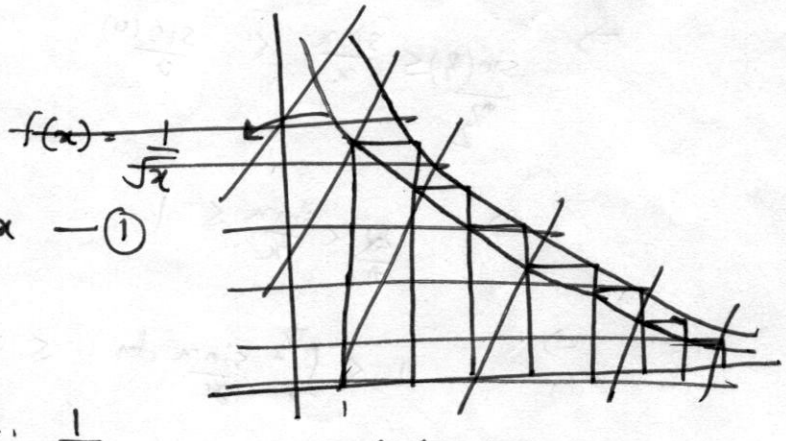
$$\Rightarrow 1 \times \pi/4 \geq \int_0^{\pi/4} \frac{\sin x}{x} \geq \frac{2\sqrt{2}}{\pi} \times \pi/4$$

$$\frac{2\sqrt{2}}{\pi} \times \pi/4 \geq \int_{\pi/4}^{\pi/2} \frac{\sin x}{x} \geq \frac{2}{\pi} \times \pi/4$$

$$\Rightarrow \frac{2\sqrt{2}+\pi}{4} \geq A \geq \frac{\sqrt{2}+1}{\sqrt{2}}$$

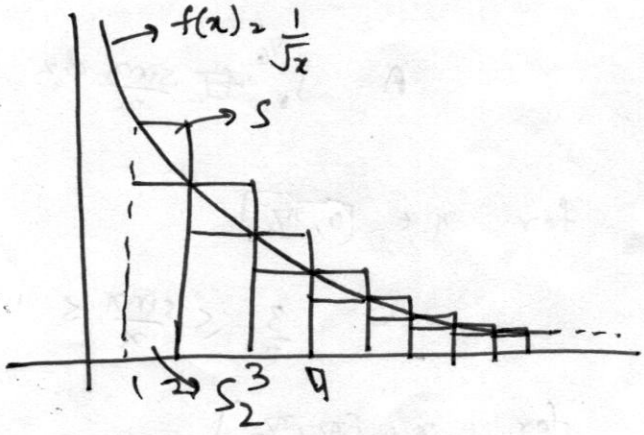


06 $S = \sum_{r=1}^{100} \frac{1}{\sqrt{r}}$



Thus $S \Rightarrow \int_1^{100} \frac{1}{\sqrt{x}} dx$ —①

$S_2 = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \frac{1}{\sqrt{4}} \cdots \frac{1}{\sqrt{101}}$



$\Rightarrow S_2 = S - \frac{1}{\sqrt{1}} + \frac{1}{\sqrt{101}}$ —③

& $S_2 < \int_1^{100} \frac{1}{\sqrt{x}} dx$ —②

from ① $S > \int_1^{100} \frac{1}{\sqrt{x}} dx$

$= 2\sqrt{x} \Big|_1^{100} = 18$

$\Rightarrow S > 18$

from ②

$S_2 < 18$

$\Rightarrow S - 1 + \frac{1}{\sqrt{101}} < 18$

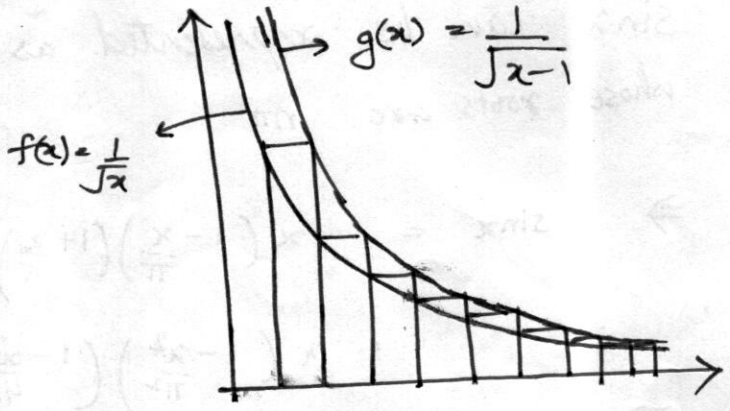
$\Rightarrow S < 19 - \frac{1}{\sqrt{101}}$

$\Rightarrow S \in (18, 19)$

$\Rightarrow [S] = 18$

$\Rightarrow \frac{1}{2}[S] = 9$

Q6



It can be seen that the required sum lies between the graph of $\frac{1}{\sqrt{x}}$ & $\frac{1}{\sqrt{x-1}}$

thus $f(x) \leq \frac{1}{\sqrt{x}} \leq g(x)$

$$\Rightarrow \int_1^{100} f(x)\,dx < \sum_{x=1}^{100} \frac{1}{\sqrt{x}} < \int_1^{100} g(x)\,dx$$

$$\Rightarrow 2\left[\sqrt{100} - 1\right] < S < 2\left[\sqrt{101} - 1\right]$$

$$\Rightarrow S > 18$$

$$\& \quad 2\left[\sqrt{101} - 1\right] < 19$$

$$\Rightarrow S < 19$$

$$\Rightarrow [S] = 18$$

$$\Rightarrow \frac{1}{2}[S] = 9$$

$\sin x$ can be represented as a polynomial whose roots are $n\pi$

$$\Rightarrow \quad \sin x = x\left(1-\frac{x}{\pi}\right)\left(1+\frac{x}{\pi}\right)\left(1-\frac{x}{2\pi}\right)\left(1+\frac{x}{2\pi}\right)\cdots$$

$$= x\left(1-\frac{x^2}{\pi^2}\right)\left(1-\frac{x^2}{4\pi^2}\right)\left(1-\frac{x^2}{9\pi^2}\right)\cdots$$

$$= x\prod_{n=1}^{\infty}\left(1-\frac{x^2}{n^2\pi^2}\right)$$

$$\Rightarrow \quad \frac{\sin x}{x} = \prod_{n=1}^{\infty} 1-\frac{x^2}{n^2\pi^2}$$

putting $x = \pm\pi/2$

$$\Rightarrow \quad \frac{1}{\pi/2} = \prod_{n=1}^{\infty}\left(1-\frac{1}{4n^2}\right)$$

$$\Rightarrow \quad \pi/2 = \prod_{n=1}^{\infty}\frac{4n^2}{4n^2-1}$$